

Problem

An AM signal is specified by $f(t) = 10(1 + 4 \cos 200\pi t) \cos \pi \times 10^4 t$

Determine the following:

(a) the carrier frequency,

(b) the lower sideband frequency,

(c) the upper sideband frequency.

Step-by-step solution

Step 1 of 2

(A) We know that the AM signal is represented by

$$f(t) = V_c [1 + V_m \cos \omega_m t] \cos \omega_c t$$

we are given

$$f(t) = 10[1 + 4 \cos 200\pi t] \cos \pi \times 10^4 t$$

comparing these two equations, we get

$$V_c = 10V; \quad \omega_m = 200\pi \text{ rad/s}; \quad \omega_c = \pi \times 10^4 \text{ rad/s}; \quad V_m = 40V$$

$$\text{so, } f_c = \frac{\omega_c}{2\pi} = \frac{\pi \times 10^4 \text{ rad/s}}{2\pi} = 5.0 \times 10^3 \text{ Hz}; \quad f_m = \frac{\omega_m}{2\pi} = \frac{200\pi \text{ rad/s}}{2\pi} = 100 \text{ Hz}$$

a.

$$\text{Carrier frequency } (f_c) = \frac{\omega_c}{2\pi} = \frac{\pi \times 10^4 \text{ rad/s}}{2\pi} = \boxed{5.0 \times 10^3 \text{ Hz}}$$

Step 2 of 2

(B) Lower side band frequency = $f_c - f_m$

$$= (5.0 \times 10^3 \text{ Hz}) - (100 \text{ Hz})$$

$$= \boxed{4.9 \times 10^3 \text{ Hz}}$$